

Hostility to Journalism, Not to the Mainstream

A citation-level audit of 278,706 posts on 4chan’s
Politically Incorrect board

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Abstract

Research on 4chan’s /pol/ board routinely frames its information environment as a contest between mainstream and alternative news media: legacy outlets are rejected, a partisan counter-canon takes their place. We test that framing on 24 daily captures spanning 45 days (2026-07-22 to 2026-09-04): 6,532 threads, 278,706 posts, 10,078 citation events over 1,408 unique domains, each carrying a transformer sentiment score for the surrounding discussion, and each domain coded against a manually built, published codebook.

The framing fails, but not in the way a null result usually fails. The mainstream–alternative contrast is flat: $p_{\text{Holm}} = 1.0$, Cliff’s $\delta = -0.045$. Crucially, the effect *shrank* as statistical power grew—manual recoding raised the alternative category from 69 to 185 citations and moved δ from -0.121 to -0.045 —which is the signature of a real null rather than an underpowered test. The three news categories (mainstream, alternative, state-controlled) are statistically indistinguishable from one another ($H = 2.97$, $p = 0.23$).

What separates cleanly is journalism from everything else. Citations to news outlets of any kind sit in markedly more negative discussion than citations to platforms, archives, institutional references, or the unclassified remainder: -0.345 against -0.183 , $p = 3.6 \times 10^{-54}$, $\delta = -0.267$, with 66.1% against 41.3% of citations in negative-scoring posts. /pol/’s hostility is organised around journalism as an activity, not around a political axis running through it. The alternative outlets supposed to constitute a counter-canon are received exactly as coldly as the BBC.

We also report that news media occupy 12.8% of citations—platforms take 43.7%, archives and file hosts 11.0%—that sourcing is extremely concentrated (Gini 0.821; YouTube alone 28.9%), and that 0.68% of posts (95% CI [0.44, 1.07]) disappear between repeated captures of the same thread, a measured floor on the survivorship bias that conditions all such corpora.

1 Introduction

4chan’s /pol/ (“Politically Incorrect”) board occupies an outsized place in accounts of online radicalisation and information disorder. Measurement work has characterised its language, its raiding behaviour, and its position in the wider web ecosystem (Hine et al., 2017; Papasavva et al., 2020). A recurring assumption in this literature, and in journalism drawing on it, is that /pol/ constitutes an *alternative media ecosystem*: a space where legacy outlets are rejected and replaced by a counter-canon of partisan publishers.

The assumption is rarely measured directly. Doing so requires knowing what /pol/ cites, in what proportion, and how the surrounding discussion treats each kind of source. This paper supplies that

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measurement for a bounded collection window, and reports that the assumption is wrong in a specific and informative way: the axis along which /pol/ sorts sources is not partisan, it is occupational.

Our contributions:

1. **A well-powered null** on the mainstream–alternative contrast, strengthened rather than weakened by added statistical power.
2. **A positive finding that replaces it**: news citations of every political stripe attract more hostile discussion than non-news citations, with the three news categories statistically homogeneous.
3. **A published codebook** of 203 manually coded domains, and evidence that automated domain lists built from US-centric rating databases undercount news on an internationally-sourced corpus by a factor of 2.8.
4. **A measured bound on survivorship bias**, from repeated captures of the same threads, in place of the usual qualitative caveat.

2 Related work

Hine et al. (2017) gave the first large-scale measurement of /pol/, characterising posting dynamics, hate-speech prevalence, and raids on other platforms. Papasavva et al. (2020) released a 3.5-year corpus of augmented /pol/ posts, establishing dataset conventions much subsequent work relies on. Both establish /pol/ as an object of quantitative study; neither takes the citation event as the unit of analysis.

Source-classification work in computational social science generally anchors labels to third-party rating databases rather than author judgment—AllSides, Media Bias/Fact Check, and Ad Fontes are common choices (Bozarth et al., 2020; Robertson et al., 2018). Section 3.3 reports what happens when that approach meets a corpus whose news citations are substantially non-American.

For sentiment we use the TweetEval RoBERTa model (Barbieri et al., 2020; Loureiro et al., 2022), whose limitations on this domain we treat as a first-order threat (Section 6).

3 Data and methods

3.1 Collection

Threads were captured from the live /pol/ JSON API via the 4TCT collector on a daily cadence, giving 24 daily captures between 2026-07-22 and 2026-09-04. This is a *snapshot* design: conclusions are scoped to the collection window, not to /pol/ in general or across time.

3.2 Citation extraction

The unit of analysis is the **citation event**: one external domain appearing in one post, deduplicated within the post. Links internal to 4chan are excluded. Mobile and shortened hosts are normalised to a canonical domain (`youtu.be`, `m.youtube.com` → `youtube.com`), and subdomain matching resolves cases such as `en.wikipedia.org`.

We flag one failure mode because it is a plausible silent defect in any such pipeline. An earlier version of ours retained only the first domain per post and discarded the rest. Because /pol/ posts characteristically lead with a video or tweet and append a source, this systematically discarded

Threads captured	6,532
Posts	278,706
Posts with ≥ 1 external link	7,107 (2.55%)
Citation events	10,078
Unique domains	1,408
Daily captures	24
Window	2026-07-22 – 2026-09-04

Table 1: Corpus summary.

the appended source: alternative-media citations were undercounted 3.6-fold. All figures here use per-citation counting.

3.3 Source coding and the undercount it corrected

Domains are coded on a medium-type axis into seven categories: mainstream, alternative, and state-controlled news; social platform; institutional reference; archive/file host; and unclassified. Deterministic structural rules handle platforms, `.gov/.mil/.edu` suffixes, and the Wikimedia family.

An initial pass using curated news lists derived from US-centric rating databases left 47.2% of citations unclassified and put news outlets at 4.6% of citations. Inspection showed that figure was an artefact. The unclassified bucket contained `dailymail.com`, `abc.net.au`, `independent.co.uk`, `itv.com`, `standard.co.uk`, `telegraph.co.uk`, `timesofisrael.com`, `aljazeera.com`, `pravda.com.ua` and many others—British, Australian, Israeli and Ukrainian outlets absent from lists built around the American media market, on a corpus dominated by threads about ongoing wars.

We therefore hand-coded the 200 most-cited unclassified domains, covering 74.9% of that bucket, and published the result as a versioned codebook of 203 domains. Four coding rules were fixed in advance and applied uniformly:

1. **Aggregators** (`msn.com`, `yahoo.com`) are coded by what they redistribute; search engines and link shorteners are not news.
2. **State-controlled** is defined by ownership and funding, not editorial independence. This places Al Jazeera, the BBC, ABC (Australia), PBS, NHK, Deutsche Welle, Anadolu, Channel News Asia and Minnesota Public Radio together, and moved the BBC (135 citations) out of mainstream.
3. **Opinion Substacks and blogs** count as alternative media: they produce editorial content.
4. **Campaign sites** are unclassified rather than a new category; state media outlets remain state-controlled.

Manual coding raised news media from 4.6% to **12.8%** of citations—a 2.8-fold undercount by the automated pass—and cut the unclassified bucket from 47.2% to 23.6%. Section 6 states what this coding does and does not license, and Section 4.5 shows why the paper’s central finding does not depend on its most contestable choice.

3.4 Sentiment measurement

Each post carrying a citation is scored with `cardiffnlp/twitter-roberta-base-sentiment-latest` (Barbieri et al., 2020; Loureiro et al., 2022); the three-way softmax collapses to a compound score $s = p_{\text{pos}} - p_{\text{neg}}$ in $[-1, +1]$. A post is scored once and the score attached to each of its citations.

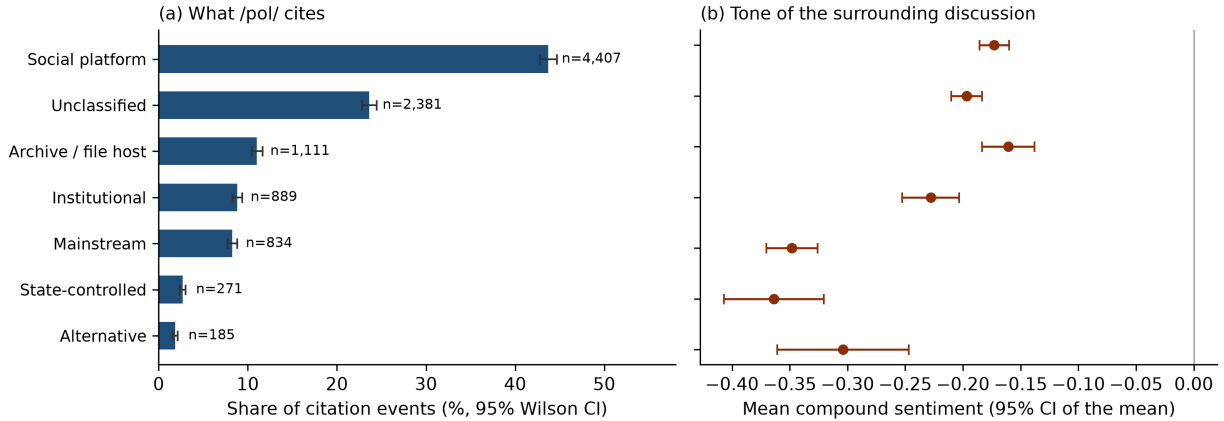


Figure 1: (a) Share of citation events by category, 95% Wilson intervals. (b) Mean compound sentiment of surrounding discussion, 95% confidence intervals *of the mean*. The three news categories cluster at left, separated from the four non-news categories at right, and overlap one another.

Category	<i>n</i>	Share	95% CI	Mean sent.	Neg. ratio
Social platform	4,407	43.7%	[42.8, 44.7]	−0.173	39.0%
Unclassified	2,381	23.6%	[22.8, 24.5]	−0.197	41.9%
Archive / file host	1,111	11.0%	[10.4, 11.7]	−0.161	45.6%
Institutional	889	8.8%	[8.3, 9.4]	−0.228	45.6%
Mainstream	834	8.3%	[7.8, 8.8]	−0.348	66.2%
State-controlled	271	2.7%	[2.4, 3.0]	−0.364	71.6%
Alternative	185	1.8%	[1.6, 2.1]	−0.304	57.8%

Table 2: Citation events by category. The rule separates news (below) from non-news (above). “Neg. ratio” is the share of citations whose surrounding post scores below -0.2 .

The score is computed over the *entire post*. It measures the **ambient tone of discussion surrounding a citation**, not the author’s stance toward the cited source. “CNN is garbage, here’s proof” scores negative and attaches to CNN even though the poster endorses the link’s content. Every tone result below must be read under that construct definition.

3.5 Statistical analysis

Proportions carry 95% Wilson score intervals (Wilson, 1927). Concentration is measured by Gini and a normalised Herfindahl–Hirschman index over per-domain citation counts. Tone uses Kruskal–Wallis (Kruskal & Wallis, 1952) as omnibus with ε^2 , then Dunn post-hoc tests with tie correction and Holm step-down adjustment (Holm, 1979); pairwise effect sizes are Cliff’s δ (Cliff, 1993) at the thresholds of Romano et al. (2006). The adjusted model regresses compound sentiment on category, thread topic, and $\log(1 + \text{post length})$, with standard errors clustered by thread. We report effect sizes throughout: at $n = 10,078$, significance is cheap.

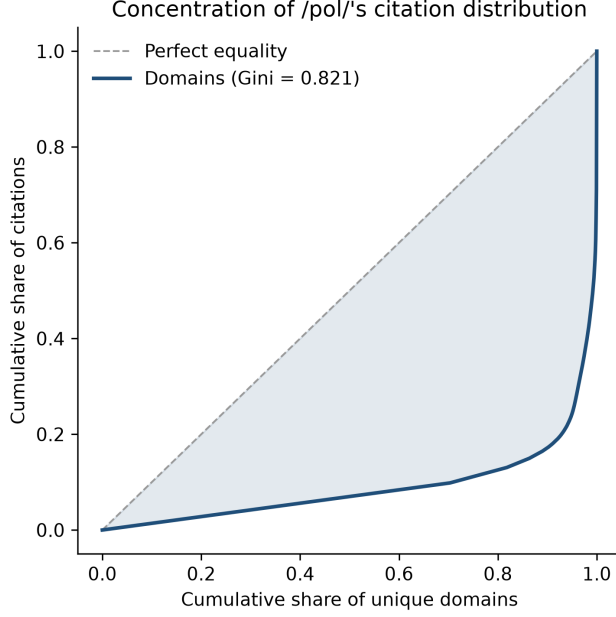


Figure 2: Lorenz curve of the citation distribution over 1,408 domains.

4 Results

4.1 What /pol/ cites

News media account for 12.8% of citation events, platforms for 43.7%, and archives and file hosts for 11.0%—the latter cited more often than any single news category. Proportions are stable across the window and show no trend: across the 24 daily captures, mainstream share ranges 3.6–12.8% (median 8.7%), platform share 37.2–53.7% (median 43.3%), and archive share 6.2–14.4% (median 11.0%).

4.2 Concentration

The diet is narrow. The domain distribution has Gini 0.821 and normalised HHI 0.091. YouTube alone supplies **28.9%** of citations; the top ten supply 49.1% and the top fifty 69.0%. Twelve domains account for half of all citations, while 989 of 1,408 domains (70.2%) appear exactly once (Figure 2).

4.3 The mainstream–alternative contrast is null

Tone differs across categories overall (Kruskal–Wallis $H = 278.0$, $p = 4.2 \times 10^{-57}$), at $\varepsilon^2 = 0.028$; the adjusted model reaches $R^2 = 0.073$.

The contrast the literature’s framing predicts is absent. Mainstream against alternative gives $z = -1.03$, $p_{\text{Holm}} = 1.0$, $\delta = -0.045$. The adjusted model agrees: relative to platforms, mainstream carries -0.104 ($p = 4.8 \times 10^{-9}$) and alternative -0.055 ($p = 0.075$), and their difference is not distinguishable.

The strength of this null rests on how it behaved under added power. Before manual recoding, alternative held 69 citations and δ was -0.121 ; that result was properly reported as underpowered. Recoding raised alternative to 185 citations—a 2.7-fold increase—and δ fell to -0.045 . An effect that *shrinks* as n grows is not an effect awaiting detection.

4.4 What separates instead: journalism from everything else

The three news categories are statistically homogeneous: Kruskal–Wallis restricted to mainstream, alternative and state-controlled gives $H = 2.97$, $p = 0.23$. None of their three pairwise contrasts survives correction (all $p_{\text{Holm}} \geq 0.88$).

Pooled against the rest of the corpus, the separation is the largest in the study:

	n	Mean	Median	Neg. ratio
News media (any kind)	1,290	−0.345	−0.374	66.1%
Everything else	8,788	−0.183	−0.054	41.3%
Mann–Whitney $p = 3.6 \times 10^{-54}$; Cliff’s $\delta = -0.267$ (small); mean gap -0.162				

Table 3: News against non-news citations.

Two thirds of news citations sit in negative-scoring posts against two fifths of everything else. The medians are more striking than the means: the typical non-news citation sits in a post scoring -0.054 , essentially neutral, while the typical news citation sits at -0.374 .

Table 4 gives the post-hoc structure that produces this. Every significant contrast involving a news category pits it against a non-news one; every contrast *within* the news block is null.

4.5 Robustness to the contestable coding choice

Our most consequential coding decision—defining state-controlled by ownership rather than editorial independence, which moves the BBC out of mainstream—is defensible but arguable. It does not threaten the central finding. Because the three news categories are statistically indistinguishable, any reallocation *among* them leaves the news/non-news partition, and therefore Table 3, unchanged. The finding is insensitive to precisely the boundary a reviewer is most likely to contest.

4.6 Archives and OSINT trackers

Two behaviours emerge that a news-centric coding scheme would hide. Archives and file hosts (`files.catbox.moe`, `reentry.org`, `archive.4plebs.org`, `archive.today`) take 11.0% of citations, more than any single news category: hosting media outside platform moderation and preserving content, including 4chan’s own deleted threads, against removal.

Within the remaining unclassified bucket, 571 citations (5.7% of the corpus, 24.0% of that bucket) across 14 domains are live war-mapping and vehicle tracking services—`liveuamap`, `deepstatemap.live`, `marinetraffic`, `flightradar24`, `adsbexchange`. On a board whose largest thread topic is armed conflict, primary-source OSINT tooling is cited about half as often as all news media combined.

4.7 Co-citation structure

Domains co-occurring within a thread form a weighted graph (edges of weight ≥ 2): 286 nodes, 1,846 edges, density 0.045, Louvain modularity 0.429 over six communities. The largest (198 domains) is general-purpose, anchored on YouTube, Wikipedia and mainstream outlets. Smaller ones are topical—a war-tracking cluster, a US-politics platform cluster (`twitter`, `truthsocial`, `rumble`), a German-language cluster. Sourcing ecosystems organise by topic and platform affordance, not by editorial orientation—consistent with the main result.

Contrast	z	p_{Holm}	δ	Magnitude
Social platform vs. Mainstream	13.55	$< 10^{-15}$	0.269	small
Archive vs. Mainstream	11.34	$< 10^{-15}$	0.305	small
Unclassified vs. Mainstream	9.70	$< 10^{-15}$	0.276	small
Social platform vs. State-controlled	8.91	$< 10^{-15}$	0.285	small
Archive vs. State-controlled	8.34	$< 10^{-15}$	0.331	medium
Institutional vs. Mainstream	7.48	1.2×10^{-12}	0.204	small
Social platform vs. Alternative	5.70	1.5×10^{-7}	0.226	small
Archive vs. Alternative	5.49	4.9×10^{-7}	0.236	small
Institutional vs. Alternative	3.42	0.0037	0.157	small
Mainstream vs. Alternative	-1.03	1.00	-0.045	negligible
Mainstream vs. State-contr.	-0.65	1.00	-0.052	negligible
State-contr. vs. Alternative	-1.36	0.88	-0.086	negligible

Table 4: Dunn contrasts with Holm correction. Above the rule: news against non-news, all significant. Below: the three within-news contrasts, all null. Non-news-versus-non-news contrasts omitted for space.

4.8 A measured bound on survivorship

Sixty-two pairs of repeated captures of the same thread let us measure what is usually asserted: **19 of 2,776 observed posts (0.68%, 95% CI [0.44, 1.07])** had disappeared by the following capture, affecting 7 threads.

This must be read for what it is. It measures post-level deletion *within threads that survived to be captured twice*. It says nothing about threads deleted entirely, nor about threads born and dying between two daily captures—which, at a daily cadence on a board with /pol/’s velocity, is the larger loss channel. It is a **floor on content loss**, from a self-selected sample, not an estimate of it. We report it because a measured floor is more useful than an unmeasured caveat.

5 Discussion

The axis is occupational, not partisan. /pol/ does not sort sources into trusted counter-media and distrusted legacy media. It sorts them into journalism and not-journalism, and treats the first with uniform coldness. Breitbart-style outlets, the supposed counter-canon, attract the same reception as the BBC; state broadcasters attract the same reception as both. Whatever generates hostility, it is not tracking the partisan position of the outlet.

Two readings are consistent with this. Under the first, the hostility is toward professional gate-keeping as such—an anti-institutional disposition that does not discriminate between institutions. Under the second, it is not about outlets at all: news links are the ones that carry contested political events, and the tone belongs to the event rather than the source. Our construct (Section 6) cannot separate these, and the adjusted model, which controls only for coarse thread topic, does not settle it. We regard disentangling them as the most valuable next step, and stance detection (Mohammad et al., 2016; Pontiki et al., 2016) as the instrument for it.

The frame is also mis-scaled. A literature asking whether /pol/ prefers alternative to mainstream media is asking about 10.1% of what /pol/ circulates, and about a distinction the board does not appear to draw. Platforms supply four times as many citations as all news combined; archives alone outnumber any single news category. Treating this as a *news* ecosystem imports a category the object does not honour.

Significance is not magnitude. Category explains under 3% of sentiment variance ($\varepsilon^2 = 0.028$; adjusted $R^2 = 0.073$). Even the news/non-news split, the strongest contrast we find, is a small effect by conventional thresholds. Post length predicts tone about as reliably as category does. These are real regularities in a noisy process, not determinants of it.

A cautionary note on automated source lists. Our first pass, using curated lists derived from US-centric rating databases, put news at 4.6% of citations. Hand-coding put it at 12.8%. The error was not random: it removed British, Australian, Israeli and Ukrainian outlets from a corpus whose dominant subject is war outside the United States. Any audit applying American media-bias taxonomies to an internationally-sourced corpus should expect an undercount of this order and check for it directly.

6 Limitations

Sentiment measurement is unvalidated on this domain. The model was trained on Twitter; /pol/’s irony, slang and toxicity are out-of-distribution, and we have *not* performed the human-annotated validation that would quantify the resulting measurement error. All tone results should be treated as provisional pending a hand-coded sample with reported model–human agreement. The descriptive, concentration and coding results do not depend on the sentiment model.

The codebook is single-coder. The 203 domains were coded by one author against fixed written rules, with uncertain cases flagged (12 of 203). No second coder annotated independently, so no Cohen’s κ or Krippendorff’s α is reported. The structural categories—platform, archive, institutional—rest on deterministic rules and are robust; the news subdivisions rest on that single-coder judgment. The central finding pools those subdivisions and is correspondingly less exposed (Section 4.5).

Coding is single-axis. A fuller design codes each domain on three orthogonal axes—medium type, editorial orientation, ownership—against external databases (AllSides, MBFC, Ad Fontes, RSF), since forcing them into one label is what makes Fox News, BBC and France 24 irreconcilable. We code medium type only, with ownership folded into the state-controlled category by rule.

A quarter of citations remain unclassified. 23.6% of citations fall outside all categories, concentrated in a long tail: 904 of the residual domains are cited exactly once. Hand-coding covered 74.9% of the original unclassified bucket; the remainder is reported as-is and could contain further news outlets, which would raise the 12.8% figure again without affecting the news/non-news contrast’s direction.

Survivorship bias is bounded below, not corrected. See Section 4.8. Moderation removes the most extreme content preferentially, so the surviving corpus is filtered in the direction of the phenomenon measured. Source-distribution estimates are relatively robust, since deletion would need to correlate with cited-source category. Tone estimates are directly exposed.

Topic classification is weak. Thread topics come from keyword matching over the opening post’s subject and slug, and 76.2% of posts fall into “Other / Misc”. The topic covariate is a coarse control, not a topical analysis—which matters for the second reading in the Discussion, since a better topic control is what would test it.

Scope. One board, 24 daily captures, 45 days. Results describe the surviving, publicly visible discourse of /pol/ during that window. They do not generalise across time or to other imageboards.

7 Conclusion

The question the field asks of /pol/—does it prefer alternative media to mainstream media?—has an answer, and the answer is that the board does not appear to make the distinction. Mainstream, alternative and state-controlled citations attract statistically indistinguishable hostility, and the null strengthened when we tripled the power available to detect a difference. What does separate is journalism from everything else: news citations of any kind sit in markedly more negative discussion than platform, archive, institutional or unclassified ones.

That reframing has a methodological consequence and a substantive one. The methodological one is that source taxonomies built for the American media market misdescribe an internationally-sourced corpus badly enough to invert a headline figure. The substantive one is that studying /pol/’s “media diet” along a left–right axis measures a distinction the board does not use. The productive next steps are target-dependent stance detection, to separate hostility toward outlets from hostility toward events, and a topic model good enough to make that separation testable.

Data and code availability

Analysis code, the manual domain codebook, derived statistics and figures are at <https://github.com/marcgarnier/4chan-pol-source-audit>. The raw capture corpus (~600 MB) is not redistributed but is reproducible from the public 4chan JSON API using the collection scripts provided. Every number in this paper is regenerated by `analysis.py` from the released database schema and codebook.

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